

Which of the following will occur during the gamma decay of an atom of technetium-99m?

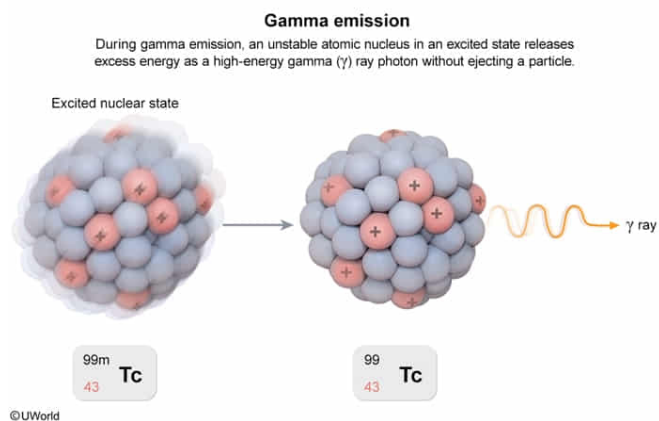
- I. The nucleus will emit a high-energy photon.
- II. The number of protons and neutrons in the nucleus will be unchanged.
- III. The atom will be converted into an isotope of a different element.

- ☐ A. I only [2%]
- ☐ B. II only [5%]
- ✓ ☒ C. I and II only [87%]
- ☐ D. I and III only [4%]

Correct

88% answered correctly

Explanation:



Radioactivity is the spontaneous **emission of radiation** (photons or particles) from an **unstable atomic nucleus**. Emitting radiation allows an unstable nucleus to achieve a more stable (lower energy) configuration (often by fragmenting into a smaller nucleus). Although several types of nuclear radiation exist, the three most common types are **alpha decay**, **beta decay**, and **gamma emission**.

During gamma emission, an unstable nucleus in an **excited state** releases excess energy by emitting a gamma ray, which is a **high-energy photon** (an **electromagnetic wave** packet) and often represented in nuclear reactions using the symbols γ or ${}^0_0\gamma$. Some radionuclides exist in a *metastable state* that is particularly prone to gamma emission. Such nuclei are labeled with the letter *m* beside the mass number within the **nuclear notation**.

One characteristic of gamma emission is that **no particle is emitted** from the nucleus, which causes the number of protons and neutrons in the nucleus to remain unchanged. As a result, the elemental identity of a nucleus is unchanged following gamma emission.

Therefore, during the gamma decay of an atom of technetium-99m, the nucleus emits a high-energy photon (**Number I**), but the number of protons and neutrons in the nucleus remains unchanged (**Number II**).

(Number III) Because the number of protons in the nucleus is unchanged, the atom will *not* be converted into an isotope of a different element.

Educational objective:

Radioactivity is the spontaneous emission of radiation (photons or particles) from an unstable atomic nucleus via common decay processes such as alpha decay, beta decay, and gamma emission. During gamma emission, a high-energy photon is released from a nucleus in an excited state without the release of a particle, which causes the elemental identity of the nucleus to remain unchanged.

Time spent:0 sec

Subject:
General Chemistry

QID:402595

System:
Atomic Theory and Chemical Composition

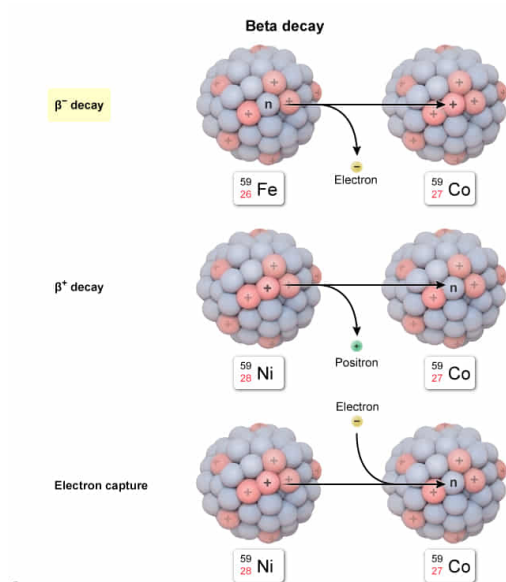
During beta decay, a radioactive atomic nucleus emits either an electron (β^- decay) or a positron (β^+ decay). If a cobalt-59 atom is formed from the β^- decay of a radionuclide, what must the original nucleus have been?

- ✓ ☒ A. $^{59}_{26}\text{Fe}$ [68%]
☐ B. $^{60}_{27}\text{Co}$ [6%]
☐ C. $^{59}_{28}\text{Ni}$ [23%]
☐ D. $^{63}_{29}\text{Cu}$ [1%]

Correct

68% answered correctly

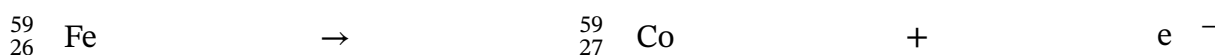
Explanation:



In radioactive nuclei, **beta decay** can occur in three forms: **β^- decay** (electron emission), **β^+ decay (positron emission)**, and electron capture. In β^- -decay, a neutron converts to a nuclear proton and emits an electron. In **β^+ decay**, a nuclear **proton converts to a neutron** (the opposite of β^- decay) and ejects a positron (an electron with a positive charge). In electron capture, a proton captures an electron near the nucleus and converts to a neutron without a positron or electron emission.

In all three forms of beta decay, the **mass number** remains **unchanged** whereas the **atomic number** either increases (β^- decay) or decreases (β^+ decay and electron capture) by 1. As the atomic number changes, the identity of the element changes accordingly.

Consequently, if an atom of ^{59}Co were formed as the result of a β^- decay, the original radionuclide would have the same mass number (59) but its atomic number would be one unit less, as shown in the following nuclear reaction:



Therefore, ^{59}Co would be formed by the β^- decay of ^{59}Fe .

(Choice B) During β^- decay, the mass number remains unchanged and the atomic number increases by one unit. ^{60}Co has a higher mass number and an unchanged atomic number, which is inconsistent with a β^- decay precursor.

(Choice C) Forming a ^{59}Co nucleus from ^{59}Ni would require ^{59}Ni to undergo β^+ decay (positron emission) or electron capture instead of the β^- decay (electron emission) specified by the question.

(Choice D) A ^{59}Co nucleus could form from ^{63}Cu only if ^{63}Cu were to undergo [alpha decay](#), but the question states that ^{59}Co was formed by β^- decay.

Educational objective:

In all forms of beta decay, the mass number remains unchanged whereas the atomic number increases (β^- decay) or decreases (β^+ decay and electron capture) by 1. β^- decay converts a neutron into a proton and emits an electron. β^+ decay and electron capture convert a proton into a neutron (the opposite of a β^- decay).

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